

॥ श्रीः ॥

लीलावती

श्रीभूतगणकचक्रचूडामणि—भास्कराचार्यविरचिता

Līlāvatī of Bhāskarācārya
(Treatise on Arithmetic, c. 1150 CE)

Critical edition from the manuscript of the
Late Pt. Manmohan Shastri Collection, Jammu
Digitized by eGangotri

Published by Shyamlal Bhat, Kashi (Varanasi)
Third Edition, Śāka 1829 (1907 CE)

Preface

The *Līlāvātī* is the first section of Bhāskara II's monumental work *Siddhāntaśiromaṇi*, composed in 1150 CE. Named after Bhāskara's daughter, it is the most celebrated textbook of Indian arithmetic, covering arithmetic operations with integers and fractions, rules of proportion, interest calculation, plane geometry, solid geometry, and combinatorial problems.

The present edition reproduces the Sanskrit verses (*śloka*) with *anvaya* (word-order analysis) and English translation, together with mathematical explanation. The source is the third edition printed at Kashi (Varanasi) by Shyamlal Bhat, preserved in the collection of the late Pandit Manmohan Shastri of Jammu.

Invocatory Verses (मङ्गलचरण)

लीलामुल्लक्षोल्कालव्यालविलसिने ।
गणेशाय नमो नीलकमलामलकान्तये ॥ १ ॥

Translation: Salutation to Gaṇeśa, the dark-lotus-hued, the playfully sportive one, having the brightness of the rising sun.

अवतार पुरा पुराणेषां यदुकुले जगतीहितेहिने ।
जयति सोऽयमिमां सकलां लीलावतीं मामपि सिद्धमहीपतिः ॥ ७ ॥

Translation: In former times, the protector of the Yadu dynasty, the benefactor of the world, incarnated. May that same victorious king complete this *Līlāvātī* together with me.

शाकाब्दे वर्षे कविवचनवती त्रिविक्रमो भूतनयोरस्य जातः ।
यो भोजराजेन कृताभिधानो विद्यापतिभूस्करभट्टनामा ॥ ८ ॥

Translation: In the Śaka year, Trivikrama was born, who was given the title Vidyāpati by King Bhoja, known as Bhūskara Bhaṭṭa.

तस्माद्विदुषदर्वज्ञो जातो गोविन्दसन्निभः ।
प्रभाकरः सुतस्तस्यात्मजस्तद्वाऽपरः ॥ ९ ॥

Translation: From him was born the all-knowing scholar Prabhākara, resembling Govinda.

तस्मान्मनोरथो जातः सतां पूर्णमनोरथः ।
श्रीमान्महेश्वराचार्यस्ततोऽजनि कवीश्वरः ॥ १० ॥

Translation: From him was born Manoratha, who fulfilled the desires of the virtuous; and from the illustrious Maheśvara Ācārya was born the lord of poets.

तत्सुतः कविबुधावनिन्दितपदः सदैव विद्यालता-
कण्ठः कविरनुप्रसादितपदः सर्वज्ञविभासदः ।
यच्छिष्यः स ह कोऽपि नो विबुधद्विद्वक्षो विबाधी क्वचित्
श्रीमान् भास्करकोविदः सममवत्सक्षितिपुण्यार्णिवतः ॥ ११ ॥

Translation: His son, a poet praised by the learned. His student was the illustrious Bhāskara, equally virtuous, ocean of good deeds.

भास्करचितयमथा सिद्धान्तशिरोमणिप्रख्याः ।
तदयं कृताऽनन्ये व्याकृतया मन्मते नियतं ॥ १५ ॥

Translation: This *Līlāvātī*, composed by Bhāskara, is the jewel on the head of the *Siddhāntaśiromaṇi*, now presented with unique explanations.

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Overview of the Līlāvātī

The *Līlāvātī* contains the following major topics:

1. मङ्गलप्रकरण (Invocatory Verses) — Salutations and genealogy
2. परिभाषा (Definitions) — Units of measurement, numerical notation
3. सङ्ख्यप्रकरण (Arithmetic) — Operations on whole numbers
4. भिन्नपरिकर्म (Fractions) — Operations with fractions
5. शून्यपरिकर्म (Zero) — Operations with zero
6. त्रैशक्ति (Rule of Three) — Direct and inverse proportion
7. मिश्रप्रकरण (Mixed Problems) — Interest, alligation, barter
8. क्षेत्रव्यवहार (Plane Geometry) — Areas, Pythagorean theorem
9. खननचर्चा (Excavations, Piles) — Volumes, stacking
10. श्रेणी (Progressions) — Arithmetic and geometric series
11. छायाव्यवहार (Shadows) — Height and distance problems
12. कुट्टक (Pulverizer) — Linear Diophantine equations
13. अङ्कपञ्चाश (Permutations) — Combinatorial problems

1 परिभाषा— Definitions and Units of Measurement

The *Līlāvātī* begins with fundamental definitions of units of measurement essential for all practical arithmetic in medieval India.

1.1 अथ शेषकम् — Units of Weight

पादोनग्रानकतुल्यष्टकैः समतुल्यैः कथितोऽत्र सेरः ॥
मणाभिधानं व्युगमैः सैरेधार्यादितोलेषु तुष्टकसङ्ख्या ॥ १ ॥

Translation: A *ser* is equal to eight *pāda* or equal to eight grains. A *maṇa* is twice the amount of a *ser* in weighing gold, etc.

द्वयकद्विसहस्रैर्घटकैः सैरस्तैः पञ्चभिः स्याद्घटिका च ताभिः ॥
मणोऽष्टमित्वालिमगीरशाहकृतास्त्र सङ्ख्या निजराज्यपृष्ठे ॥ २ ॥

Translation: Two thousand *ghaṭakas* make a *ser*; five of these *ser*s make a *ghaṭikā*. The weight of one *maṇa* in this kingdom follows this system.

1.2 कालपरिभाषा— Definitions of Time

शोभाः कालादिपरिभाषा लोकतः प्रसिद्धा देया ॥

Translation: The definitions of time and other measures are well-known in the world.

The traditional Indian system of time measurement: 60 *vipalas* = 1 *pala*; 60 *palas* = 1 *ghaṭikā* (24 minutes); 60 *ghaṭikās* = 1 day and night; 15 days = 1 *pakṣa* (fortnight); 2 *pakṣas* = 1 month; 12 months = 1 year.

2 सङ्ख्यप्रकरण — Arithmetic Operations

This chapter covers the fundamental operations: addition, subtraction, multiplication, division, squaring, square roots, cubing, and cube roots.

2.1 सङ्कलतिव्यकलति — Addition and Subtraction

सूत्रः यथोक्तं परिकल्प्य गणितं कुर्यात्।

As taught, having set up the numbers, one should compute.

Numbers must be aligned by place value before performing addition or subtraction. The commentary demonstrates this with worked examples.

Example: Add 2, 5, 32, 193, 18, 10, and 100.

Set up and sum: $2 + 5 + 32 + 193 + 18 + 10 + 100 = 360$.

Example: Subtract 10000 from 360: $10000 - 360 = 9640$.

2.2 गुणकारणसूत्र — Multiplication

सूत्र: समद्विघातः कृतिः उच्यते

A product of two equal numbers is called a square (kṛti).

Example: To multiply 135 by 12:

$$135 \times 12 = 135 \times 10 + 135 \times 2 = 1350 + 270 = 1620$$

Bhāskara gives several methods: direct multiplication, breaking into parts (distributive property), and using algebraic identities.

2.3 भागकारणसूत्र — Division

Division is performed by repeated subtraction. The commentary gives detailed worked examples following the place-value method.

Example: $360 \div 12 = 30$.

2.4 वर्गक्रियणसूत्र — Squaring

सूत्र: समद्विघातः कृतिः उच्यते

The product of two equal numbers is called a square.

Methods for squaring a number:

Method 1: Direct multiplication $n \times n$

Method 2: Using the formula $(a + b)^2 = a^2 + 2ab + b^2$

Method 3: Using the formula $n^2 = (n + a)(n - a) + a^2$

सूत्र: खण्डाभ्यां वा हतो राशिरिष्टः खण्डचतुष्कयुक् ॥

Or, the number multiplied by the component parts of itself, combined with the square of the difference...

2.5 वर्गमूलकरणसूत्र — Square Root Extraction

सूत्र: अथ स्थाप्योऽन्त्यवर्गो द्विगुणान्त्यनिघ्नः ॥ स्यादपरिच्छेदः तथा परेऽभ्यस्तयुक्तवान्त्यमपि राशिम् ॥ ८ ॥

Set down the number. Find the largest square in the leftmost group. This gives the first digit of the root. Subtract its square, bring down the next group, divide by twice the current root, and repeat.

Example: Find $\sqrt{65536}$:

$\sqrt{65536}$	256
4	(largest square ≤ 6)
255	bring down 55
$45 \times 5 = 225$	$2 \times 20 = 40$, $255 \div 40 \approx 5$
3036	bring down 36
$506 \times 6 = 3036$	$2 \times 250 = 500$, $3036 \div 500 \approx 6$
0	remainder zero

Therefore $\sqrt{65536} = 256$.

2.6 घनकरणसूत्र — Cubing

सूत्र: समत्रिघातः घनः उच्यते

The product of a number multiplied three times by itself is called a cube.

Methods for cubing:

Method 1: Direct multiplication $n \times n \times n$

Method 2: Using $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

2.7 घनमूलकरणसूत्र — Cube Root Extraction

The method follows a similar place-value decomposition as square root extraction, but using cubes of digits.

Example: $\sqrt[3]{1728} = 12$, since $12^3 = 1728$.

3 भन्निपरकिर्म — Operations with Fractions

3.1 जतचितुष्टय — Four Classes of Fractions

Bhāskara defines four types of fractions:

1. भन्नि (bhinna) — Simple fraction $\frac{a}{b}$
2. प्रभा (prabhāga) — Part of a part
3. भगामुबन्ध (bhāgānubandha) — Fraction combined with an integer
4. भगामवह (bhāgāpavāha) — Fraction less than an integer

3.2 भन्निगुणनभगारसूत्र — Multiplication and Division of Fractions

सूत्र: नरघनरके क्रमशो गुणयेत् तद् भागहारकैः ॥

Multiply the numerator by the numerator and the denominator by the denominator for multiplication; for division, invert and multiply.

Example: $\frac{2}{3} \times \frac{5}{7} = \frac{2 \times 5}{3 \times 7} = \frac{10}{21}$

Example: $\frac{2}{3} \div \frac{5}{7} = \frac{2}{3} \times \frac{7}{5} = \frac{14}{15}$

3.3 भन्निवर्गघनसूत्र — Squares and Cubes of Fractions

सूत्र: वर्गोघनौ च वर्गघनकारकृतौ भिन्नस्य ॥

The square and cube of a fraction are obtained by squaring or cubing its numerator and denominator.

Example: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$, $\left(\frac{2}{3}\right)^3 = \frac{8}{27}$

4 शून्यपरकिर्म — Operations with Zero

सूत्र: योगे खं क्षयसङ्ख्ये व्यतिकले शून्यं व्येके गुणके हारे ।
खं खं गुणं गुणकारे शून्यं युक्तं च युक्तावियुक्तं च ॥

Translation: In addition, zero makes the number unchanged; in subtraction, zero makes the number unchanged; in multiplication, the product is zero; when zero is the divisor, the result is *khahāra* (a fraction with zero denominator, i.e., infinity).

Bhāskara gives the following rules for operations involving *śūnya* (zero):

- $a + 0 = a$ (addition)
- $a - 0 = a$ (subtraction)
- $a \times 0 = 0$ (multiplication)
- $0 \times 0 = 0$ (multiplication of zero by zero)
- $\frac{0}{a} = 0$ (division of zero by a number)
- $\frac{a}{0}$ is *kha-hāra* (a fraction with zero denominator), considered infinite

सूत्र: खहरं तत्र गुण्यं यदि षुण्यं तत् समं गुण्यस्य ॥

In *khahāra*, if zero is the multiplier, the result equals the multiplicand.

Bhāskara's concept of *khahāra* (a fraction with zero denominator) is remarkable — he treats it as representing infinity, and notes that adding or subtracting any finite quantity from it leaves it unchanged.

असङ्ख्याः खहरा निर्दिष्टा असंख्यैः गुणकैः ॥

Infinite *khahāra* quantities are described, multiplied by infinite multipliers...

5 त्रैरशकि — The Rule of Three

The Rule of Three (*trairāśika*) is one of the most important topics in Indian arithmetic. It solves problems of the form:

$$\text{“If } A \text{ gives } B, \text{ what does } C \text{ give?”} \quad \text{Answer: } \frac{B \times C}{A}$$

5.1 इच्छकर्मप्रकार — Direct Rule of Three

सूत्र: प्रमाणफलमिच्छाफलं चेत् कुर्यात्।

Multiply the fruit by the desire and divide by the measure.

Example: If 5 books cost 60 rupees, what do 8 books cost?

$$\text{Price} = \frac{60 \times 8}{5} = 96 \text{ rupees}$$

5.2 व्यस्तत्रैरशकि — Inverse Rule of Three

सूत्र: व्यस्ते विपरीतं कुर्यात्।

In the inverse case, do the reverse.

When the relationship is inverse (more workers take less time, etc.):

$$\text{Result} = \frac{B \times A}{C}$$

Example: If 8 workers complete a job in 15 days, how many days will 12 workers take?

$$\text{Days} = \frac{8 \times 15}{12} = 10 \text{ days}$$

5.3 पञ्चरशकिद्वकि — Extended Rules

सूत्र: पञ्चराशिके मध्यमफलं द्विधा कृत्वा ॥

In the five-fold rule, divide the middle fruit into two parts...

Extended rules for compound proportions:

- पञ्चरशकि (Five-fold rule): Two intermediate terms
- सप्तरशकि (Seven-fold rule): Three intermediate terms
- नवरशकि (Nine-fold rule): Four intermediate terms

6 मशिरप्रकरण — Mixed Problems

This chapter contains a variety of practical problems covering interest, barter, alligation, and other commercial arithmetic.

6.1 शैलरशकिवधि — Alligation

सूत्र: मिश्रमूल्यात्प्रत्येकमूल्यं विशोध्य शुद्धिकं प्रति विक्रयेण ॥

Subtract each individual price from the mixed price; the differences give the proportions for exchange.

Problem: If rice of quality A costs 8 rupees/kg and rice of quality B costs 12 rupees/kg, in what ratio should they be mixed to get a mixture worth 10 rupees/kg?

$$\text{Ratio} = (12 - 10) : (10 - 8) = 2 : 2 = 1 : 1$$

6.2 व्याजवर्धन — Compound Interest

सूत्र: मूलं व्याजेन वर्धितं पुनस्तेन वर्धितं यदि स्यात् ॥

The principal increased by interest, then again increased by that same interest...

Problem: 100 rupees lent at 5% per month for 1 year. Find the amount.

$$A = P(1 + r)^n = 100 \times \left(\frac{21}{20}\right)^{12}$$

6.3 वसिष्ठप्रकरण — Filling of Tanks

Problem: A tank has three pipes. The first can fill it in 5 hours, the second in 6 hours, and the third can empty it in 4 hours. If all three are opened together, how long will it take to fill the tank?

$$\text{Net fill rate} = \frac{1}{5} + \frac{1}{6} - \frac{1}{4} = \frac{12 + 10 - 15}{60} = \frac{7}{60}$$

$$\text{Time to fill} = \frac{60}{7} = 8\frac{4}{7} \text{ hours.}$$

7 क्षेत्रव्यवहार — Plane Geometry

क्षेत्रे व्यवहारः ---

This section deals with the calculation of areas of various plane figures and the properties of right triangles.

7.1 आयतक्षेत्र — Rectangle

सूत्रः दैर्घ्यं विस्तारेण गुणितं क्षेत्रफलं भवति।

The area of a rectangle is length multiplied by breadth.

7.2 भुजकोटकिर्णनयन — The Pythagorean Theorem

सूत्रः भुजे द्वादशके यो यो कोटिकर्णविनिर्णये कथा ॥

प्रकाराभ्यां वद विप्र तो तावकर्णी गतौ ॥ ५ ॥

Translation: O learned Brahmin! The sides (bhujā) being 12, tell me quickly the hypotenuse (karṇa) and perpendicular (koṭi) by both methods.

सूत्रः इष्टयोराहतिर्द्वयोः कोटिवर्गान्तरं भुजः ॥

कृतियोगस्तयोरेवं कर्णस्त्वकरणीगतः ॥ ८ ॥

Translation and Mathematical Explanation:

This verse gives the relationship between the sides of a right triangle:

- **Base** (भुज, *bhujā*): The horizontal side
- **Perpendicular** (कोटि, *koṭi*): The vertical side
- **Hypotenuse** (कर्ण, *karṇa*): The diagonal side

The rule states:

$$\text{karṇa}^2 = \text{bhujā}^2 + \text{koṭi}^2$$

This is the famous Pythagorean theorem, known in India as the *Śulba-sūtra* theorem.

Example: Find the hypotenuse when base = 3 and perpendicular = 4:

$$\text{karṇa} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Example from the text: For base = 12, perpendicular = 5:

$$\text{karṇa} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Example: For base = 9, perpendicular = 12:

$$\text{karṇa} = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$$

7.3 कोटिजिज्ञप्ति — Finding the Koṭi from Bhujā and Karṇa

When the hypotenuse and one side are known:

$$\text{koṭi} = \sqrt{\text{karṇa}^2 - \text{bhujā}^2}$$

Example: karṇa = 85, bhujā = 36:

$$\text{koṭi} = \sqrt{85^2 - 36^2} = \sqrt{7225 - 1296} = \sqrt{5929} = 77$$

Verification:

$$36^2 + 77^2 = 1296 + 5929 = 7225 = 85^2 \checkmark$$

7.4 चतुर्भुजक्षेत्र — Quadrilateral Figures

सूत्र: चतुर्भुजस्य क्षेत्रफलं व्यासक्षेत्रं वा सामान्यम्।

The area of a quadrilateral: the product of the diagonals divided by 2.

Rectangle: Area = length $\times$ breadth

Square: Area = side²

Trapezium: Area = $\frac{1}{2} \times$ (sum of parallel sides) $\times$ height

Rhombus: Area = $\frac{1}{2} \times$ product of diagonals

7.5 वृत्तक्षेत्र — Circle

सूत्र: व्यासवर्गपादं भुजवर्गस्य पञ्चगुणस्य चतुर्थांशः।

वृत्तफलं व्यासार्धगुणं वा ॥

The area of a circle:

$$A = \frac{d^2}{4} \times \pi \approx \frac{d^2}{4} \times \frac{22}{7} = \frac{11d^2}{14}$$

Or using the modern form: $A = \pi r^2$ where $r = d/2$.

Bhāskara uses $\pi \approx \frac{22}{7}$ (the same approximation used in the *Āryabhaṭīya*).

7.6 त्रिभुजक्षेत्र — Triangle

सूत्र: षड्भिर्भुजवर्गैः कोटिवर्गैश्चतुर्भिः सहितैः ॥

क्षेत्रफलं त्रिभुजस्य स्यात्तद्वर्गमूलं द्विगुणितं वा ॥

Heron's formula: Given sides a, b, c , compute $s = \frac{a+b+c}{2}$:

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

Example: Sides 13, 14, 15: $s = 21$, $A = \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$.

8 खत्तचतिशिरेण्यद्द — Excavations, Piles, and Progressions

8.1 खत्तव्यवहार — Excavations

खातव्यवहारः ---

This section deals with calculations related to excavations, trenches, and tanks.

Problem: An excavation is to be made with length 30, breadth 20, and depth 8. Find the volume of earth to be removed.

$$V = l \times b \times d = 30 \times 20 \times 8 = 4800 \text{ cubic units}$$

For excavations with sloping sides, the *śāṇi* (average) method is used:

$$V = \frac{A_1 + A_2 + A_3 + A_4}{4} \times \text{length}$$

where A_1, A_2, A_3, A_4 are the cross-sectional areas at different depths.

8.2 चतिव्यवहार — Piles of Bricks

This section deals with stacking problems and the calculation of volumes of piles.

Problem: Bricks are piled in the form of a pyramid with a rectangular base. The base has 10 bricks along one side and 6 along the other. The pile is 5 layers high. Find the total number of bricks.

$$N = 10 \cdot 6 + 9 \cdot 5 + 8 \cdot 4 + 7 \cdot 3 + 6 \cdot 2 = 60 + 45 + 32 + 21 + 12 = 170$$

For square pyramids, the formula is:

$$N = \frac{n(n+1)(2n+1)}{6}$$

8.3 श्रेणी— Progressions

श्रेण्यांसा ---

The *śreṇī* (progression) section covers arithmetic and geometric progressions.

समश्रेणी— Arithmetic Progression

सूत्र: आदिरुत्तरगुणितः पूर्विका सङ्कलितो भवति ॥

The first term multiplied by the common difference, combined with the sum of the preceding terms...

For an arithmetic progression with first term a , common difference d , and n terms:

- n -th term: $a_n = a + (n - 1)d$
- Sum of n terms: $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$

where l is the last term.

Example: Find the sum of the first 100 natural numbers.

$$S_{100} = \frac{100}{2}(1 + 100) = 50 \times 101 = 5050$$

गुणोत्तरश्रेणी— Geometric Progression

सूत्र: गुणोत्तरे गुणकसङ्गुणिते पूर्वसमासः ॥

In a geometric progression, multiply the first term by the ratio raised to the power of the number of terms...

For a geometric progression with first term a and common ratio r :

- n -th term: $a_n = a \cdot r^{n-1}$
- Sum of n terms: $S_n = a \cdot \frac{r^n - 1}{r - 1}$

Problem: A king offers a reward of 1 grain of rice on the first square of a chessboard, 2 on the second, 4 on the third, and so on, doubling each time. Find the total grains on all 64 squares.

$$S_{64} = 1 + 2 + 4 + \dots + 2^{63} = \frac{2^{64} - 1}{2 - 1} = 2^{64} - 1 = 18446744073709551615$$

9 छायाव्यवहार — Shadow Problems

छायाव्यवहारकथन ---

This chapter uses the properties of similar triangles to solve problems involving shadows, heights of objects, and distances.

9.1 दीप्तच्छाया— Finding the Height of a Lamp

सूत्र: छायाग्रदूरं द्विजोत्कर्षं कुर्यात्।

The distance to the tip of the shadow multiplied by the height of the object, divided by the length of the shadow, gives the height of the light source.

Problem: A lamp is placed on a pillar. A man of height 6 stands 8 units away from the pillar, and his shadow extends 12 units from his feet. Find the height of the lamp.

Solution: Using similar triangles:

$$\frac{h}{6} = \frac{8 + 12}{12} = \frac{20}{12}$$

$$h = 6 \times \frac{20}{12} = 10 \text{ units}$$

9.2 द्वच्छिमा— Difference of Two Shadows

When an object casts different shadows from two different light sources:

$$\frac{h_1}{s_1} = \frac{h_2}{s_2} = \frac{d}{s_1 - s_2}$$

10 कुट्टकव्यवहार — The Pulverizer

कुट्टकव्यवहार: ---

The *kutṭaka* (pulverizer) is the method for solving linear Diophantine equations of the form:

$$ax + by = c$$

where a , b , and c are given integers, and x , y are unknown integers to be found.

10.1 स्थरिकुट्टक — Fixed Pulverizer

सूत्र: द्विच्छेदः कुट्टकः---

The pulverizer is the division of two (numbers).

The method uses the Euclidean algorithm to find the greatest common divisor, then back-substitutes to find the solution.

Problem: Find x and y such that $121x + 17 = y$

Solution: Using the *kutṭaka* method:

Step 1: Apply the pulverizer to 121 and 17:

$$\begin{aligned} 121 &= 7 \times 17 + 2 \\ 17 &= 8 \times 2 + 1 \end{aligned}$$

Step 2: Work backwards to express 1 as a combination of 121 and 17:

$$\begin{aligned} 1 &= 17 - 8 \times 2 \\ &= 17 - 8 \times (121 - 7 \times 17) \\ &= 17 - 8 \times 121 + 56 \times 17 \\ &= 57 \times 17 - 8 \times 121 \end{aligned}$$

So $x = 8$, $y = 121 \times 8 + 17 = 985$ gives a solution.

10.2 सङ्क्रमणप्रकार — Transposition Method

The transposition method converts a given equation into the standard form by rearrangement.

11 अङ्कपत्रा — Combinatorial Problems

अङ्कपाशः ---

This chapter contains problems related to permutations and combinations, and number puzzles.

11.1 अङ्कपत्रावधि — Digit Arrangements

सूत्र: अङ्कानां क्रमविपर्ययैर्बहुधा योगः प्रदर्श्यते ॥

The sum (of numbers formed) by the permutations of digits in all orders is demonstrated.

Problem: Using the digits 1, 2, 3, 4, how many different 4-digit numbers can be formed?

Solution: $4! = 24$ different numbers.

Problem: Find the sum of all numbers formed by permuting the digits 2, 5, 6, 7.

Solution: Each digit appears in each place $3! = 6$ times. Sum of digits = 20. Total sum = $6 \times 20 \times (1 + 10 + 100 + 1000) = 120 \times 1111 = 133320$.

Famous Problems from the Līlāvātī

The *Līlāvātī* is justly famous for its poetically stated problems. Bhāskara presents mathematical puzzles in beautifully composed Sanskrit verses, often set in natural or domestic scenes.

The Bee Problem (मधुकरगणतिम्)

आसां पञ्चनमुनीनां कुटजशैलीन्द्रक्रमेण त्रयो भृङ्गाः ।
पञ्चमः पुनरागत्य तिष्ठत्येको मधुकरी मुखे ॥

पञ्चनमुनीनां तृतीयांशं शिलीन्द्रं तथापरे त्रयः ।
नवम एकश्च द्रुतं वेगात् कुटजमूले तिष्ठति ॥

Translation: One-fifth of a swarm of bees settled on a *kuṭaja* (*Wrightia antidysenterica*) flower, one-third on a *śilīndhra* (*Bassia latifolia*) flower. Three times the difference between these two numbers flew over a *krāta* (a scarlet flower). One bee was left hovering about, attracted by the fragrance of a jasmine and a pandanus. Tell me, charming lady, how many bees were there?

Solution: Let x be the total number of bees.

$$\frac{x}{5} + \frac{x}{3} + 3\left(\frac{x}{3} - \frac{x}{5}\right) + 1 = x$$

$$\frac{x}{5} + \frac{x}{3} + 3 \cdot \frac{2x}{15} + 1 = x$$

$$\frac{3x + 5x + 6x}{15} + 1 = x \implies \frac{14x}{15} + 1 = x \implies x = 15$$

Answer: 15 bees.

The Lotus Problem (पद्मसमस्या)

पद्मस्य पृष्ठे तरुणी किंचित्प्रलम्बमानं दृष्ट्वा हृष्टा ।
उद्धृत्य करेण तलं प्राप्तं वायुना पातितं दूरात् ॥

उदकाद्विप्रोन्नतस्य तोये लीलावती दत्तमित्यादि ।
उच्चस्तम्भस्योच्छ्रायं वद त्वं तत्त्वतो यदि जानासि ॥

Translation: Beautiful Līlāvātī, a lotus flower stands a units above the water surface. When blown by the wind, it sinks d units from its original vertical position and touches the water. Find the depth of the water and the length of the lotus stem.

Mathematical formulation: If the lotus is a units above the water, the stem length equals the depth h plus a . When blown, the stem forms the hypotenuse of a right triangle:

$$(h + a)^2 = h^2 + d^2$$

$$h^2 + 2ha + a^2 = h^2 + d^2$$

$$h = \frac{d^2 - a^2}{2a}$$

Example: If the lotus stands 1 *āṅgula* above the water and is blown 2 *hastas* sideways:

$$h = \frac{2^2 - (1/24)^2}{2 \times (1/24)} \approx \frac{4}{1/12} = 48 \text{ āṅgulas}$$

The Pearl Necklace Problem (मुक्ताहारसमस्या)

मुक्ताहारः स्तनान्तःस्थितो मणिः श्रेण्यां पतन् वेगात् ।
स्तनभूमिं प्राप्य त्रिभागं श्रेण्याः प्रचक्रमे पद्मे ॥

Translation: A string of pearls with a jewel at the center was hanging down. The necklace broke and the pearls scattered. One-third of the pearls fell to the ground, one-fifth remained on the couch, one-sixth was saved by the lady, one-tenth was caught by her lover, and six pearls remained on the string. Tell me the total number of pearls in the necklace.

Solution: Let x be the total number of pearls.

The fraction accounted for is:

$$\frac{1}{3} + \frac{1}{5} + \frac{1}{6} + \frac{1}{10} = \frac{10 + 6 + 5 + 3}{30} = \frac{24}{30} = \frac{4}{5}$$

The remaining pearls = $x - \frac{4x}{5} = \frac{x}{5} = 6$, so $x = 30$.

Answer: 30 pearls.

The Peacock and the Snake (मयूरसर्पसमस्या)

मयूरोऽङ्गनकूटे स्थितो बिलात्सर्पो दूरतः पश्यन् ।
पतत्यरेखां कृत्वा यावत् सर्पो विलं गतस्तावत् ॥

Translation: A peacock is perched on the top of a pillar. A snake is at the base of the pillar, at some distance from it. The peacock pounces on the snake in a straight line, covering equal distances down the pillar and along the ground. Find the point where they meet.

Mathematical formulation: Let the pillar height be P and the snake's distance from the base be D . The peacock flies in a straight line to meet the snake. If the meeting point is at distance x from the base:

$$P - x = \sqrt{x^2 + D^2}$$

Squaring both sides:

$$P^2 - 2Px + x^2 = x^2 + D^2$$

$$x = \frac{P^2 - D^2}{2P}$$

Example: Pillar height = 9, snake's distance = 27:

$$x = \frac{81 - 729}{18} = \frac{-648}{18} = -36$$

This negative value indicates the formulation; the actual problem as stated in the *Līlāvātī* has the peacock flying along the hypotenuse.

The Swallow Problem (हंससमस्या)

Problem: Swallows are sitting on the branches of a tree. If each branch has 3 swallows, one swallow is left over. If each branch has 4 swallows, one branch is left empty. How many swallows and branches are there?

Solution: Let s = number of swallows, b = number of branches.

$$s = 3b + 1$$

$$s = 4(b - 1) = 4b - 4$$

Equating: $3b + 1 = 4b - 4$, so $b = 5$ and $s = 16$.

Answer: 16 swallows and 5 branches.

The Elephant and the Shepherd (गजरक्षकसमस्या)

गजानां षोडशदन्तैर्मिथोऽभिमुखयोर्द्वयोः संयुक्तयोः ।
दन्तानां सङ्ख्यां वद त्वं यदि गणिते कुशलोऽसि ॥

Translation: Two elephants, each having 16 tusks (8 on each side), face each other in battle. If their tusks become interlocked, tell me the total number of tusks.

Solution: A simple counting problem: each elephant has 16 tusks, so together they have $16 + 16 = 32$ tusks. The problem tests the student's ability to extract relevant information from a poetic description.

The Snake and the Peacock (सर्पमयूरक्रीडा)

सर्पोऽप्यङ्गनकूटे मयूरं दृष्ट्वा तरसा पतति ।
यावदन्तर्भूमौ गच्छति तावत्पतत्यपि स मयूरः ॥

Translation: A peacock is sitting on a pillar of height 9 *hastas*. A snake is at the base of the pillar, at a distance of 27 *hastas*. Seeing the snake, the peacock pounces in a straight line. Meanwhile, the snake also moves toward the pillar. Find the distance from the pillar where they meet.

Solution: This is a classic pursuit problem. If the peacock flies along the hypotenuse and the snake crawls along the ground, meeting at distance x from the pillar:

$$\text{Peacock's path} = \sqrt{9^2 + (27 - x)^2}$$

Assuming equal speeds, they meet where the peacock's flight distance equals the snake's remaining distance. Setting up the equation and solving gives the meeting point.

Appendix: Sanskrit Mathematical Vocabulary

Numbers and Operations

Sanskrit	Transliteration	Meaning
सङ्ख्या	<i>saṅkhyā</i>	number
अङ्क	<i>anka</i>	digit
शून्य	<i>śūnya</i>	zero
एक	<i>eka</i>	one
दश	<i>daśa</i>	ten
शत	<i>śata</i>	hundred
सहस्र	<i>sahasra</i>	thousand
कोटि	<i>koṭi</i>	ten million
योग	<i>yoga</i>	addition
व्यवकलन	<i>vyavakalana</i>	subtraction
गुणन	<i>guṇana</i>	multiplication
भागहार	<i>bhāgahāra</i>	division
वर्ग	<i>varga</i>	square
वर्गमूल	<i>vargamūla</i>	square root
घन	<i>ghana</i>	cube
घनमूल	<i>ghanamūla</i>	cube root
भिन्न	<i>bhinna</i>	fraction
रूप	<i>rūpa</i>	integer (unit)
प्रमाण	<i>pramāṇa</i>	argument (in rule of three)
फल	<i>phala</i>	result
इच्छा	<i>icchā</i>	requisition

Geometric Terms

Sanskrit	Transliteration	Meaning
क्षेत्र	<i>kṣetra</i>	plane figure, field
भुजा	<i>bhujā</i>	side, base
कोटि	<i>koṭi</i>	perpendicular
कर्ण	<i>karna</i>	hypotenuse, diagonal
व्यास	<i>vyāsa</i>	diameter
व्यासार्ध	<i>vyāsārdha</i>	radius
परिधि	<i>paridhi</i>	circumference
दैर्घ्य	<i>dairghya</i>	length
विस्तार	<i>vistāra</i>	breadth

Sanskrit	Transliteration	Meaning
ऊन	<i>ūna</i>	less, diminished
अधिक	<i>adhika</i>	more, increased
क्षेत्रफल	<i>kṣetraphala</i>	area
घनफल	<i>ghanaphala</i>	volume

Colophon

Līlāvātī of Bhāskarācārya
 Critical edition prepared from the manuscript preserved in the
Late Pt. Manmohan Shastri Collection, Jammu
 Digitized by eGangotri
 Original printed edition:
 Published by **Shyamlal Bhat**, Kashi (Varanasi)
 Third Edition, Śāka 1829 (1907 CE)

*“Līlāvātī has been the most beloved of all mathematical textbooks in India.
 Named after Bhāskara’s daughter, it has instructed generations
 of students in the beautiful art of calculation.”*